

Combinatorics of (p, q) -analogues of bi-Stirling Eulerian polynomials

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Outline

1. Eulerian polynomials: background, permutation interpretation
2. Bi-Stirling Eulerian polynomials: connection formula, γ -positivity
3. (p, q) -analogues of bi-Stirling Eulerian polynomials: Flajolet's Lemma on continued fraction, Françon-Viennot bijection, γ -positivity
4. Bi-Stirling-Euler-Mahonian polynomials: remixed Eulerian numbers
5. Link to Cluster parking functions

Eulerian polynomials: from Euler's definition

Consider the following two linear operators on $\mathbb{C}[x]$:

X : multiplication by x ;
 D : the usual derivative.



Leonhard Euler (1707-1783)

Euler's definition

$$(D \cdot X)^n \left(\frac{1}{1-x} \right) = \sum_{i \geq 0} (i+1)^n x^i = \frac{A_n(x)}{(1-x)^{n+1}}.$$

$$\sum_{i \geq 0} x^i = \frac{1}{1-x}$$

$$\sum_{i \geq 0} (i+1)x^i = \frac{1}{(1-x)^2}$$

$$\sum_{i \geq 0} (i+1)^2 x^i = \frac{1+x}{(1-x)^3}$$

$$\sum_{i \geq 0} (i+1)^3 x^i = \frac{1+4x+x^2}{(1-x)^4}$$

Euler's exponential function formula

$$\sum_{n \geq 0} A_n(x) \frac{z^n}{n!} = \frac{(1-x)e^{z(1-x)}}{1-xe^{z(1-x)}}.$$

Permutation

Let $[n] := \{1, 2, \dots, n\}$. The *symmetric group* $\mathfrak{S}_n$ is the set of all bijections $\sigma : [n] \rightarrow [n]$. Elements of $\mathfrak{S}_n$ are called *permutations* of $[n]$. We use two ways to write a permutation $\sigma \in \mathfrak{S}_n$.

Example

If $\sigma \in \mathfrak{S}_5$ is defined by $\sigma(1) = 2$, $\sigma(2) = 4$, $\sigma(3) = 5$, $\sigma(4) = 1$, and $\sigma(5) = 3$, then we have

Linear: $\sigma = 24513$ Cyclic: $\sigma = (1\ 2\ 4)(3\ 5)$.

Permutation statistics: a mapping $\text{st} : \mathfrak{S}_n \rightarrow \mathbb{N}$

We first recall two classical *Eulerian statistics*. For a permutation $\sigma \in \mathfrak{S}_n$, an index $1 \leq i \leq n-1$ is called

- a *descent* of σ if $\sigma(i) > \sigma(i+1)$;
- an *excedance* of σ if $\sigma(i) > i$.

Denote by $\text{des}(\sigma)$ and $\text{exc}(\sigma)$ the number of descents and excedances of σ .

Example

If $\sigma = 54132 = (1\ 5\ 2\ 4\ 3) \in \mathfrak{S}_5$, then $\text{des}(\sigma) = |\{1, 2, 4\}| = 3$, $\text{exc}(\sigma) = |\{1, 2\}| = 2$.

Permutation statistics: a mapping $\text{st} : \mathfrak{S}_n \rightarrow \mathbb{N}$

Eulerian polynomials have the following combinatorial interpretations:

$$A_n(x) = \sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle x^k = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)},$$

where $\left\langle n \atop k \right\rangle := |\{\sigma \in \mathfrak{S}_n : \text{des}(\sigma) = k \text{ or } \text{exc}(\sigma) = k\}|$ are called the *Eulerian numbers*.

- MacMahon (1905) proved that des and exc are equidistributed;
- Carlitz and Riordan (1955) showed equals $A_n(x)$ (in case des);
- Foata (1965) introduced a bijection proving that des and exc are equidistributed.

Foata's fundamental transformation

Let $\sigma = 4271365 \in \mathfrak{S}_7$. Then $\sigma = (1\ 4)(2)(3\ 7\ 5)(6)$.

Reversing the entries in each cycle and writing the smallest entry at the end gives $\hat{\sigma} = (4\ 1)(2)(5\ 7\ 3)(6)$. Erasing the parentheses, we obtain

$$\Phi(\sigma) = \underline{4}1257\underline{3}6.$$

Moreover, $\text{exc}(\sigma) = |\{1, 3\}| = 2$ and $\text{des}(\Phi(\sigma)) = |\{1, 5\}| = 2$.

Foata's fundamental transformation

An entry π_i is a *right-to-left minimum* (**rmin**) of π if

$$\pi_i < \pi_j \quad \text{for all } j > i.$$

Conversely, given $\pi = \Phi(\sigma)$, we recover the cycles of σ by cutting π immediately after each right-to-left minimum.

$$4125736 = 41 \mid 2 \mid 573 \mid 6 \longmapsto (41)(2)(573)(6) \longmapsto (14)(2)(375)(6).$$

Hence

$$\text{cyc}(\sigma) = \text{rmin}(\Phi(\sigma)).$$

Stirling statistics via Foata's transformation

The number of permutations in $\mathfrak{S}_n$ with k cycles is the Stirling number of the first kind:

$$c(n, k) = \#\{\sigma \in \mathfrak{S}_n : \text{cyc}(\sigma) = k\} = \#\{\pi \in \mathfrak{S}_n : \text{rmin}(\pi) = k\}.$$

Different choices of cycle arrangement lead to the four statistics:

smallest at the end, increasing order	$\longleftrightarrow$	$\text{rmin},$
smallest at the beginning, decreasing order	$\longleftrightarrow$	left-to-right minimum (lmin),
largest at the beginning, increasing order	$\longleftrightarrow$	left-to-right maximum (lmax),
largest at the end, decreasing order	$\longleftrightarrow$	right-to-left maximum (rmax).

These statistics are called the *Stirling statistics*.

Some properties of polynomials

A polynomial $f(x) = \sum_{i=0}^n a_i x^i \in \mathbb{R}[x]$ is called

- *symmetric*, with the center of symmetry $n/2$, if $a_i = a_{n-i}$ for all $0 \leq i \leq n$;
- *unimodal*, if $0 \leq a_0 \leq a_1 \leq \dots \leq a_k \geq a_{k+1} \geq \dots$ for some $0 \leq k \leq n$;
- *γ -positive*, if it can be written as

$$f(x) = \sum_{i=0}^{\lfloor n/2 \rfloor} \gamma_i x^i (1+x)^{n-2i}, \quad \gamma_i \geq 0.$$

Unimodality is a central property in combinatorics. Moreover, for polynomials with nonnegative coefficients,

$$\gamma\text{-positive} \implies \text{symmetric and unimodal}.$$

Gamma-positivity of Eulerian polynomials

For $\sigma \in \mathfrak{S}_n$, with the convention $\sigma_0 = \sigma_{n+1} = 0$, a letter $\sigma_i \in [n]$ is called a *double descent* (**dd**) of σ if $\sigma_{i-1} > \sigma_i > \sigma_{i+1}$.

Foata and Schützenberger (1970) proved the following elegant γ -positive expansion of the Eulerian polynomials

$$A_n(x) = \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \gamma_{n,k} x^k (1+x)^{n-1-2k},$$

where $\gamma_{n,k} := |\{\sigma \in \mathfrak{S}_n : \text{des}(\sigma) = k, \text{dd}(\sigma) = 0\}|$.

Gamma-positivity of Eulerian polynomials

The first γ -expansions of $A_n(x)$ are as follows,

$$A_n(x) = \begin{cases} 1 + x, & \text{if } n = 2, \\ (1 + x)^2 + 2x, & \text{if } n = 3, \\ (1 + x)^3 + 8x(1 + x), & \text{if } n = 4, \\ (1 + x)^4 + 22x(1 + x)^2 + 16x^2, & \text{if } n = 5, \\ (1 + x)^5 + 52x(1 + x)^3 + 136x^2(1 + x), & \text{if } n = 6. \end{cases}$$

In a follow-up work, [Foata and Schützenberger \(1973\)](#) showed that the normalized γ -coefficients $\gamma_{n,k}/2^k$ count the André permutations with k descents.

Gamma-positivity of binomial-Eulerian polynomials

Postnikov, Reiner, and Williams (2008) define the *binomial-Eulerian polynomials*,

$$\tilde{A}_n(x) := 1 + x \sum_{m=1}^n \binom{n}{m} A_m(x) = \sum_{\sigma \in \mathcal{M}_{n+1}} x^{\text{des}(\sigma)},$$

where $\mathcal{M}_n$ is the set of permutations $\sigma \in \mathfrak{S}_n$ such that the first descent (if any) of π appears at the letter n .

$$\mathcal{M}_2 = \{12, 21\} \quad \text{and} \quad \mathcal{M}_3 = \{123, 132, 231, 312, 321\}.$$

Gamma-positivity of binomial-Eulerian polynomials

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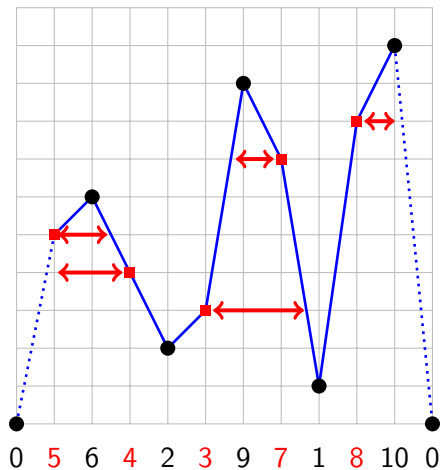
$$\mathcal{M}_2 = \{12, 21\} \quad \text{and} \quad \mathcal{M}_3 = \{123, 132, 231, 312, 321\}.$$

They proved the following γ -expansion

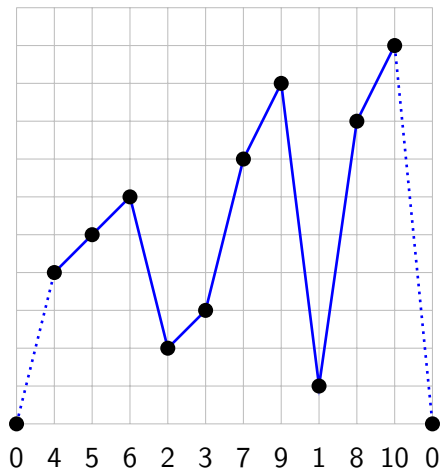
$$\tilde{A}_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \tilde{\gamma}_{n,k} x^k (1+x)^{n-2k},$$

where $\tilde{\gamma}_{n,k} := |\{\sigma \in \mathcal{M}_{n+1} : \text{des}(\sigma) = k, \text{dd}(\sigma) = 0\}|$.

Valley-hopping method (duality)



representative



Idea of the proof

Valley-hopping partitions $\mathfrak{S}_n$ into orbits.

For each orbit, choose the unique representative with no double descents:

$$\sigma \mapsto \hat{\sigma}, \quad \text{dd}(\hat{\sigma}) = 0.$$

If $\hat{\sigma}$ has k descents and m movable double ascents, then

$$\sum_{\tau \in \text{Orb}(\hat{\sigma})} x^{\text{des}(\tau)} = x^k (1+x)^m.$$

Therefore,

$$A_n(x) = \sum_{\hat{\sigma} : \text{dd}(\hat{\sigma})=0} x^{\text{des}(\hat{\sigma})} (1+x)^{\text{da}(\hat{\sigma})} = \sum_k \gamma_{n,k} x^k (1+x)^{n-1-2k}.$$

$$\gamma_{n,k} = |\{\hat{\sigma} \in \mathfrak{S}_n : \text{dd}(\hat{\sigma}) = 0, \text{des}(\hat{\sigma}) = k\}|.$$

Cycle (α, t) -Eulerian polynomials

An index $i \in [n]$ is called a

- *drop* (**drop**) of σ if $i > \sigma(i)$;
- *fixed point* (**fix**) of σ if $i = \sigma(i)$.

Carlitz and Scoville (1977) defined the *cycle (α, t) -Eulerian polynomials*,

$$A_n^{\text{cyc}}(x, y, t \mid \alpha) := \sum_{\sigma \in \mathfrak{S}_n} x^{\text{exc}(\sigma)} y^{\text{drop}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)},$$

where $\text{cyc}(\sigma)$ denotes the number of cycles of σ .

Cycle (α, t) -Eulerian polynomials

The exponential generating function of polynomials $A_n^{\text{cyc}}(x, y, t | \alpha)$ is given by

Theorem (Carlitz and Scoville 1977)

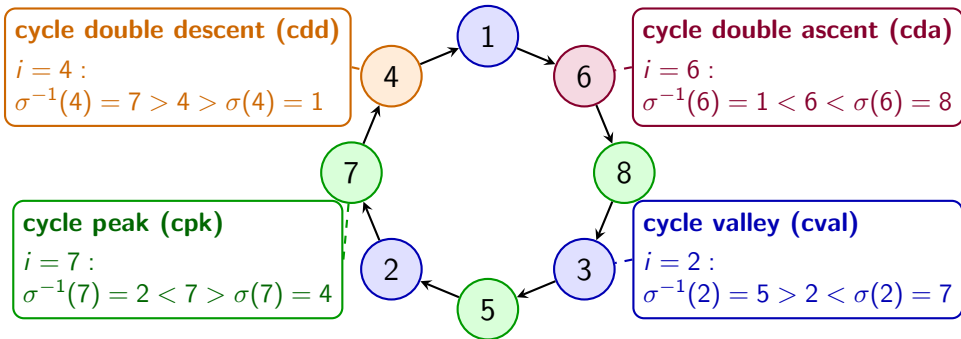
We have

$$\sum_{n \geq 0} A_n^{\text{cyc}}(x, y, t | \alpha) \frac{z^n}{n!} = \left(\frac{(x - y)e^{tz}}{xe^{yz} - ye^{xz}} \right)^\alpha.$$

History: This identity was implicitly established by **Foata–Schützenberger** (1970) using the exponential formula; **Brenti** (2000) gave a proof using symmetric functions and plethysm.

Cyclic conditions

$$\sigma = (1\ 6\ 8\ 3\ 5\ 2\ 7\ 4) = 67512843.$$



For this example: $\text{cval} = \text{cpk} = 3$, $\text{exc} = \text{cval} + \text{cda} = 4$, $\text{drop} = \text{cpk} + \text{cdd} = 4$.

Refinement of cycle (α, t) -Eulerian polynomials

Theorem (Zeng, 1993)

If $xy = u_1 u_2$ and $x + y = u_3 + u_4$, then

$$1 + \sum_{n \geq 1} \frac{z^n}{n!} \sum_{\sigma \in \mathfrak{S}_n} (u_1 u_2)^{\text{cpk}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} = \left(\frac{(x - y)e^{tz}}{xe^{yz} - ye^{xz}} \right)^\alpha.$$

Comparing this formula and the generating functions of $A_n^{\text{cyc}}(x, y, t \mid \alpha)$, we have the following expansion.

A refined cycle (α, t) -Eulerian expansion

If $xy = u_1 u_2$ and $x + y = u_3 + u_4$, then

$$A_n^{\text{cyc}}(x, y, t \mid \alpha) = \sum_{\sigma \in \mathfrak{S}_n} (u_1 u_2)^{\text{cpk}(\sigma)} u_3^{\text{cda}(\sigma)} u_4^{\text{cdd}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)}.$$

We give a combinatorial proof using [cyclic valley-hopping](#), which combines Foata's transformation and valley-hopping, namely, $\Phi \circ \text{Valley-hopping} \circ \Phi^{-1}$.

bi-Stirling Eulerian polynomials

Carlitz and Scoville (1974) defined the *bi-Stirling Eulerian polynomials*,

$$A_n(x, y \mid \alpha, \beta) := \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{asc}(\sigma)} y^{\text{des}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1},$$

where $\text{asc}(\sigma)$ is the number of ascents of $\sigma \in \mathfrak{S}_{n+1}$, namely,

$$\text{asc}(\sigma) = |\{ i \in [n] : \sigma_i < \sigma_{i+1} \}|.$$

Theorem (Carlitz and Scoville (1974))

$$\sum_{n \geq 0} A_n(x, y \mid \alpha, \beta) \frac{z^n}{n!} = (1 + xF(x, y; z))^\alpha (1 + yF(x, y; z))^\beta,$$

where $F(x, y; z) = \frac{e^{xz} - e^{yz}}{xe^{yz} - ye^{xz}}$.

Refinement of bi-Stirling Eulerian polynomials

For $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$ with $\sigma_0 = \sigma_{n+1} = \mathbf{0}$, the index $i \in [n]$ is called a

- *valley* (**val**) of σ if $\sigma_{i-1} > \sigma_i < \sigma_{i+1}$; *peak* (**pk**) of σ if $\sigma_{i-1} < \sigma_i > \sigma_{i+1}$;
- *double ascent* (**da**) of σ if $\sigma_{i-1} < \sigma_i < \sigma_{i+1}$;

Obviously, we have the following identities:

Linear conditions: $\text{val} = \text{pk} - 1$, $\text{asc} = \text{val} + \text{da}$, $\text{des} = \text{val} + \text{dd}$.

Refinement of bi-Stirling Eulerian polynomials

For $\sigma = \sigma_1 \dots \sigma_n \in \mathfrak{S}_n$ with $\sigma_0 = \sigma_{n+1} = 0$, the index $i \in [n]$ is called a

- **valley (val)** of σ if $\sigma_{i-1} > \sigma_i < \sigma_{i+1}$; **peak (pk)** of σ if $\sigma_{i-1} < \sigma_i > \sigma_{i+1}$;
- **double ascent (da)** of σ if $\sigma_{i-1} < \sigma_i < \sigma_{i+1}$;

Obviously, we have the following identities:

Linear conditions: $\text{val} = \text{pk} - 1$, $\text{asc} = \text{val} + \text{da}$, $\text{des} = \text{val} + \text{dd}$.

Ji (2023) considered the following polynomials. Let $\mathbf{u} = (u_1, u_2, u_3, u_4)$.

$$A_n(\mathbf{u} \mid \alpha, \beta) := \sum_{\sigma \in \mathfrak{S}_{n+1}} (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}.$$

Theorem (Ji)

$$\sum_{n \geq 0} A_n(\mathbf{u} \mid \alpha, \beta) \frac{z^n}{n!} = (1 + xF(x, y; z))^{\frac{\alpha+\beta}{2}} (1 + yF(x, y; z))^{\frac{\alpha+\beta}{2}} e^{\frac{1}{2}(\beta-\alpha)(u_3-u_4)z},$$

where $x + y = u_3 + u_4$, $xy = u_1 u_2$ and $F(x, y; t) = \frac{e^{xt} - e^{yt}}{x e^{yt} - y e^{xt}}$.

Generalized Eulerian polynomials

We use the following combined local-record statistics: an index $i \in [n]$ is a

statistic	local shape	Stirling property
<i>left-to-right-maximum-peak</i> (lmaxpk)	peak	left-to-right maximum
<i>right-to-left-maximum-peak</i> (rmaxpk)	peak	right-to-left maximum
<i>left-to-right-maximum-da</i> (lmaxda)	double ascent	left-to-right maximum
<i>right-to-left-maximum-dd</i> (rmaxdd)	double descent	right-to-left maximum

Generalized Eulerian polynomials

Define the *generalized bi-Stirling Eulerian polynomials*

$$A_n(\mathbf{u}, f, g, t \mid \alpha, \beta) = \sum_{\sigma \in \mathfrak{S}_{n+1}} (u_1 u_2)^{\text{val}(\sigma)} u_3^{\text{da}(\sigma)} u_4^{\text{dd}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} \\ \times t^{\text{lmaxda}(\sigma)+\text{rmaxdd}(\sigma)} \alpha^{\text{lmax}(\sigma)-1} \beta^{\text{rmax}(\sigma)-1}.$$

- When $(f, g, h) = (1, 1, 1)$, it reduces to Ji's polynomial;
- When $\mathbf{u} = (1, x, 1, x)$ and $(f, g, h) = (1, 1, 1)$, it reduces to Carlitz and Scoville's polynomial;
- When $\mathbf{u} = (1, x, 1, x)$, $(\alpha, \beta) = (1, 1)$, and $(f, g, h) = (1, 1, 1)$, it reduces to the Eulerian polynomial $A_n(x)$;
- When $\mathbf{u} = (1, x, 1, x)$, $(\alpha, \beta) = (1, 1)$, and $(f, g, h) = (0, 1, 1)$, it reduces to the binomial-Eulerian polynomial $\tilde{A}_n(x)$. (Here $f = 0$ forces n to appear in the first descent position.)

A connection formula

Theorem (Xu and Zeng)

If $xy = u_1 u_2$ and $x + y = u_3 + u_4$, then

$$A_n^{\text{cyc}} \left(x, y, \frac{\alpha u_3 + \beta u_4}{\alpha f + \beta g} t \mid \alpha f + \beta g \right) = A_n(\mathbf{u}, f, g, t \mid \alpha, \beta).$$

Taking $\beta = 0$, the sum over $\mathfrak{S}_{n+1}$ reduces to one over $\mathfrak{S}_n$. Under the specialization $u_2 = u_4 = x$ and $u_1 = u_3 = \alpha = f = g = t = 1$, the formula recovers the classical equidistribution of des and exc.

We construct a bijection based on the expansion of cycle (α, t) -Eulerian polynomials and two variants of [Foata's fundamental transformation](#).

Sketch of the proof

From the expansion of cycle (α, t) -Eulerian polynomials, we have

$$\begin{aligned} A_n^{\text{cyc}}(x, y, t(\alpha u_3 + \beta u_4)(\alpha f + \beta g)^{-1} \mid \alpha f + \beta g) \\ = \sum_{\pi \in \mathfrak{S}_n} (u_1 u_2)^{\text{cpk}(\pi)} u_3^{\text{cda}(\pi)} u_4^{\text{cdd}(\pi)} (t(\alpha u_3 + \beta u_4))^{\text{fix}(\pi)} (\alpha f + \beta g)^{\text{cyc}(\pi) - \text{fix}(\pi)}. \end{aligned}$$

Then, using the binomial theorem, this can be translated into coloring the cycles of π red or blue.

$$(\alpha + \beta)^{|E|} = \sum_{h: E \rightarrow \{\textcolor{red}{R}, \textcolor{blue}{B}\}} \alpha^{\#\{x \in E : h(x) = \textcolor{red}{R}\}} \beta^{\#\{x \in E : h(x) = \textcolor{blue}{B}\}}.$$

Sketch of the proof

Let $\pi = (157)(364)(2)(8)(9) \in \mathfrak{S}_9$. Then

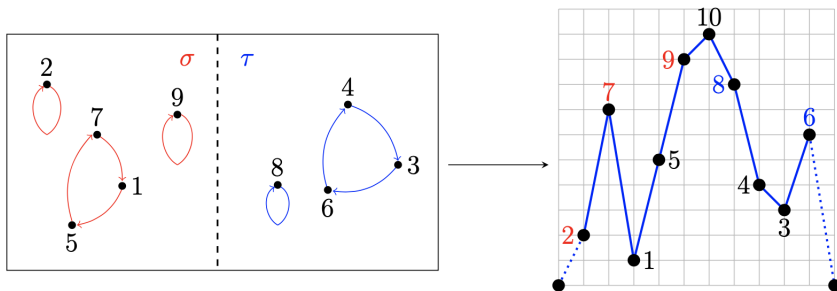


FIGURE 2. The mapping $\rho : (\sigma, \tau) \mapsto \tilde{\pi} = \mathbf{27159108436}$.

Gamma-positivity

Setting $\alpha = \beta$ and $g = 1$ in the connection formula (with conditions $xy = u_1 u_2$ and $x + y = u_3 + u_4$), we have

$$A_n(\mathbf{u}, f, 1, t \mid \alpha, \alpha) = A_n^{\text{cyc}} \left(x, y, \frac{x+y}{f+1} t \mid \alpha(f+1) \right).$$

Setting $\mathbf{u} = (1, x, 1+x, 0)$ and using the **Linear conditions**: $\text{asc} = \text{val} + \text{da}, \dots$ and the **Cyclic conditions**: $\text{exc} = \text{cval} + \text{cda}, \dots$, we have

Lemma

$$\begin{aligned} A_n(\mathbf{u}, f, 1, t \mid \alpha, \alpha) &= \sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \text{dd}(\sigma)=0}} x^{\text{des}(\sigma)} (1+x)^{n-2\text{des}(\sigma)} f^{\text{lmaxpk}(\sigma)-1} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2} \\ &= \sum_{\substack{\sigma \in \mathfrak{S}_n \\ \text{cdd}(\sigma)=0}} x^{\text{drop}(\sigma)} (1+x)^{n-2\text{drop}(\sigma)} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)} (f+1)^{\text{cyc}(\sigma)-\text{fix}(\sigma)}. \end{aligned}$$

Gamma-positivity

Let $\mathbf{u} = (1, x, 1, x)$. Then

$$A_n(x, t, f \mid \alpha) := A_n(1, x, 1, x, f, 1, t \mid \alpha, \alpha).$$

By the definition of $A_n(\mathbf{u}, f, g, t \mid \alpha, \beta)$, we have

$$A_n(x, t, f \mid \alpha) := \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{des}(\sigma)} t^{\text{lmaxda}(\sigma) + \text{rmaxdd}(\sigma)} f^{\text{lmaxpk}(\sigma) - 1} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2},$$

Then, collecting terms according to the powers of x , the preceding lemma yields the following expansion.

Gamma-positivity

Theorem (Xu and Zeng)

For $0 \leq j \leq \lfloor n/2 \rfloor$, we have

$$A_n(x, t, f | \alpha) = \sum_{j=0}^{\lfloor n/2 \rfloor} \gamma_{n,j}(\alpha, t, f) x^j (1+x)^{n-2j},$$

where

$$\begin{aligned} \gamma_{n,j}(\alpha, t, f) &= \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=j}^{\text{dd}=0}} \alpha^{\text{lmax}(\sigma) + \text{rmax}(\sigma) - 2} t^{\text{rmaxdd}(\sigma)} f^{\text{lmaxpk}(\sigma) - 1} \\ &= \sum_{\sigma \in \mathfrak{S}_{n, \text{drop}=j}^{\text{cdd}=0}} \alpha^{\text{cyc}(\sigma)} t^{\text{fix}(\sigma)} (f+1)^{\text{cyc}(\sigma) - \text{fix}(\sigma)}. \end{aligned}$$

This result unifies, refines and extends (from linear setting to cyclic setting) the γ -expansions of the Eulerian polynomial and the binomial-Eulerian polynomials (in the case $f = 0$).

Table: γ -coefficients with $f = 1$

For convenience, let $A_n(x, t | \alpha) := A_n(x, t, 1 | \alpha)$ and $\gamma_{n,j}(\alpha, t) := \gamma_{n,j}(\alpha, t, 1)$.

$n \setminus j$	0	1	2
0	1		
1	αt		
2	$\alpha^2 t^2$	2α	
3	$\alpha^3 t^3$	$2\alpha + 6\alpha^2 t$	
4	$\alpha^4 t^4$	$2\alpha + 8\alpha^2 t + 12\alpha^3 t^2$	$4\alpha + 12\alpha^2$
5	$\alpha^5 t^5$	$2\alpha + 10\alpha^2 t + 20\alpha^3 t^2 + 20\alpha^4 t^3$	$16\alpha + 40\alpha^2 + 20\alpha^2 t + 60\alpha^3 t$

Table: The first values of $\gamma_{n,j}(\alpha, t)$ for $0 \leq 2j < n \leq 5$.

For $0 \leq j \leq \lfloor n/2 \rfloor$, let $d_{n,j}(\alpha, t) = \gamma_{n,j}(\alpha, t)/2^j$, then,

$$A_n(x, t | \alpha) = \sum_{j=0}^{\lfloor n/2 \rfloor} 2^j d_{n,j}(\alpha, t) x^j (1+x)^{n-2j}.$$

André permutations

For each x , write

$$\sigma = w_1 w_2 x w_4 w_5,$$

where w_2, w_4 are the maximal adjacent blocks with letters $> x$.

André condition

$$\text{dd}(\sigma) = 0, \quad w_4 = \emptyset \Rightarrow w_2 = \emptyset, \quad \max(w_2) < \max(w_4)$$

whenever $w_2, w_4 \neq \emptyset$.

Example at $x = 1$

213 : 2 **1** 3, $2 < 3 \implies$ André permutation,

312 : 3 **1** 2, $3 > 2 \implies$ not André permutation.

Cycle André permutations

Let $\mathcal{A}_n^1$ be the set of André permutations of the first kind in $\mathfrak{S}_n$. It is known that the cardinality of $\mathcal{A}_n^1$ is the *Euler number* E_n , which is the n -th coefficient in

$$\sum_{n \geq 0} E_n \frac{z^n}{n!} = \sec(z) + \tan(z).$$

A cycle $C = (a_1, \dots, a_k)$ is called an *André cycle* if $a_1 = \min\{a_1, \dots, a_k\}$ and the word $a_2 \dots a_k$ is an André permutation of the first kind. We say that a permutation is a *cycle André permutation* if it is a product of disjoint André cycles. Let $\mathcal{CA}_n$ be the set of cycle André permutations of $[n]$.

Cycle André permutations

Example

The permutation

$$\sigma = (1, 2, 3)(4, 5)$$

is a cycle André permutation. Indeed, both cycles are André cycles:

$$1 = \min\{1, 2, 3\}, \quad 4 = \min\{4, 5\},$$

and the words 23 and 5 are André permutations of the first kind.

On the other hand,

$$\tau = (1, 5, 2, 3)(4, 6)$$

is not a cycle André permutation, since the cycle $(1, 5, 2, 3)$ is not an André cycle, 523 (order-isomorphic to 312) is not an André permutation of the first kind.

Combinatorial interpretation for $d_{n,j}(\alpha, t)$

Theorem (Xu and Zeng)

For $0 \leq j \leq \lfloor n/2 \rfloor$, we have

$$d_{n,j}(\alpha, t) = \sum_{\substack{\sigma \in \mathcal{CA}_n \\ \text{drop}(\sigma)=j}} t^{\text{fix}(\sigma)} \alpha^{\text{cyc}(\sigma)}, \quad (1)$$

$$d_{n,j}(\alpha, t) = \sum_{\substack{\sigma \in \mathcal{A}_{n+1}^{(i)} \\ \text{des}(\sigma)=j}} t^{\text{rmin}(\sigma)} \alpha^{\text{rmin}(\sigma)-1}, \quad (i = 1, 2). \quad (2)$$

We prove this theorem by computing the exponential generating functions of both sides, and derive the $i = 1$ case of (2) from (1) by a bijection from $\mathcal{CA}_n$ to $\mathcal{A}_{n+1}^1$, and the $i = 2$ case by constructing another bijection from $\mathcal{A}_{n+1}^1$ to $\mathcal{A}_{n+1}^2$ via the so-called André trees model, where $\mathcal{A}_{n+1}^2$ is the set of André permutations of the second kind.

(p, q) -analogues of bi-Stirling Eulerian polynomials

Motzkin path, continued fractions, Françon-Viennot bijection

(Weighted) Motzkin paths

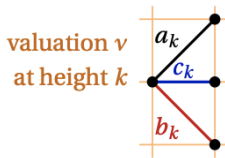
A Motzkin path is a lattice path ω starting at $(0,0)$, ending at $(n,0)$, and never going below the x -axis, with steps

$$\nearrow, \quad \rightarrow, \quad \searrow.$$

The height of the i -th step is

$$h_i = \#\{j < i : w_j = \nearrow\} - \#\{j < i : w_j = \searrow\}.$$

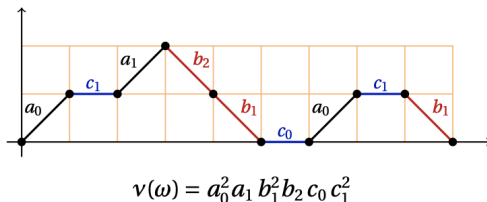
A step starting at height k has weight



The weight of a path ω is

$$v(\omega) = \prod_{\text{steps } s \text{ of } \omega} v(s).$$

Flajolet's Fundamental Lemma on Jacobi-continued fractions



Lemma (Flajolet 1980)

We have

$$\sum_{n \geq 0} \sum_{\omega \in \mathcal{M}_n} \nu(\omega) z^n = \mathcal{J}(z; \mathbf{c}, \boldsymbol{\lambda}) := \frac{1}{1 - c_0 z - \frac{a_0 b_1 z^2}{1 - c_1 z - \frac{a_1 b_2 z^2}{\ddots}}},$$

where $\mathbf{c} := (c_k)_{k \geq 0}$ and $\boldsymbol{\lambda} := (\lambda_k)_{k \geq 1}$ with $\lambda_k = a_{k-1} b_k$ be elements of a commutative ring.

Laguerre histories

A **2-Motzkin path** is a Motzkin path whose horizontal steps come in two colors:

 and .

Definition (Laguerre history)

A Laguerre history of length n is a pair

$$(w, p), \quad p = (p_1, \dots, p_n),$$

where w is a 2-Motzkin path and

$$0 \leq p_i \leq h_i \quad (1 \leq i \leq n).$$

We denote by $\mathcal{L}_n$ the set of Laguerre histories of length n .

Françon–Viennot bijection

For $\sigma \in \mathfrak{S}_{n+1}$, Françon–Viennot's bijection gives

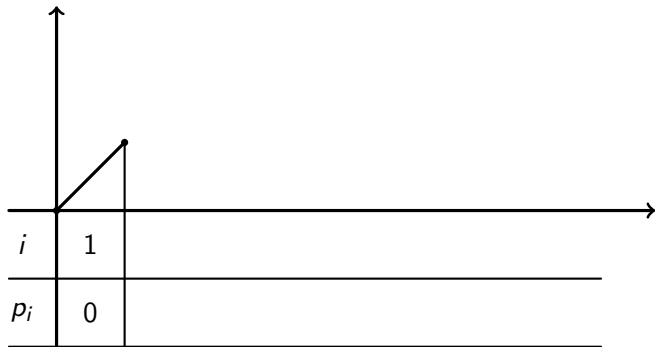
$$\Psi_{FV}(\sigma) = (w, p) \in \mathcal{L}_n, \quad w = w_1 \cdots w_n, \quad p = (p_1, \dots, p_n).$$

The step w_i is determined by the local shape of i in σ :

local shape of i	w_i
valley	
peak	
double descent	
double ascent	

The label is given by $p_i = (3i-2)_i(\sigma)$.

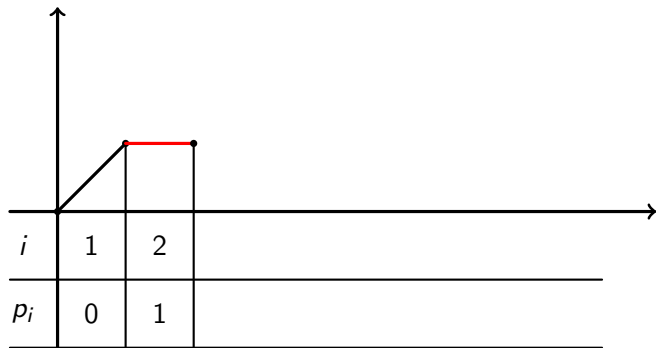
Example: $\Psi_{FV}(\sigma)$, where $\sigma = 697835142$.



Step 1. For $i = 1$. Since $5 > 1 < 4$, 1 is a *valley*, so $w_1 = \nearrow$. There is no descent to the left of 1 which contains 1. Hence

$$p_1 = (31 - 2)_1(\sigma) = 0.$$

Example: $\Psi_{FV}(\sigma)$, where $\sigma = 697835142$.

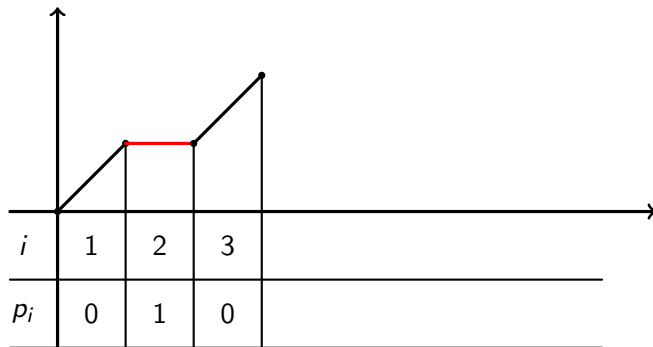


Step 2. For $i = 2$. Since $4 > 2 > 0$, 2 is a *double descent*, so $w_2 = \rightarrow$. To compute p_2 , we look at descents to the left of 2. We have

$$5 > 2 > 1.$$

Thus $p_2 = (31 - 2)_2(\sigma) = 1$.

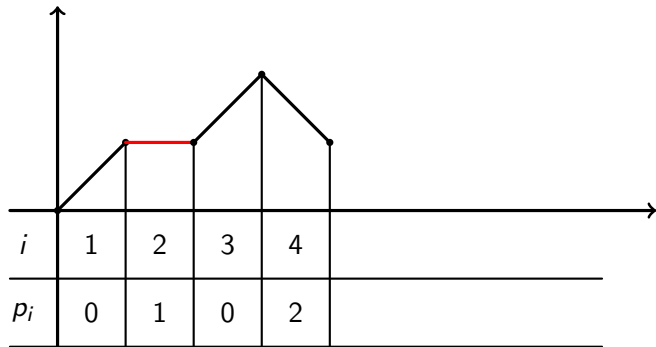
Example: $\Psi_{FV}(\sigma)$, where $\sigma = 697835142$.



Step 3. For $i = 3$. Since $8 > 3 < 5$, 3 is a *valley*, so $w_3 = \nearrow$.
 There is no descent pair to the left of 3 which contains 3. Hence

$$p_3 = (31 - 2)_3(\sigma) = 0.$$

Example: $\Psi_{FV}(\sigma)$, where $\sigma = 697835142$.

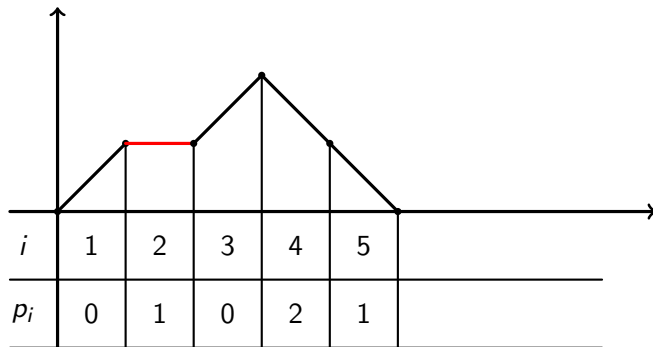


Step 4. For $i = 4$. Since $1 < 4 > 2$, 4 is a *peak*, so $w_4 = \searrow$. There are two descent pair to the left of 4 which contain 4:

$$8 > 4 > 3, \quad 5 > 4 > 1.$$

Thus $p_4 = (31 - 2)_4(\sigma) = 2$.

Example: $\Psi_{FV}(\sigma)$, where $\sigma = 697835142$.

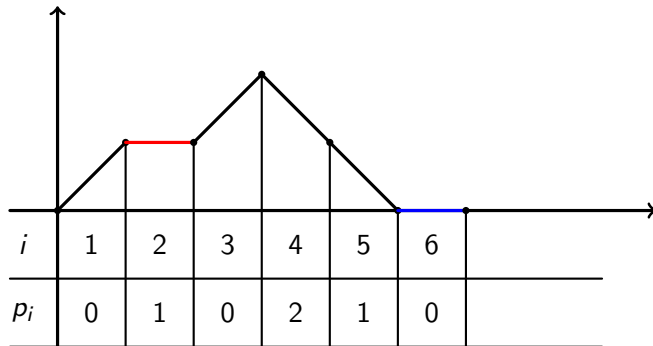


Step 5. For $i = 5$. Since $3 < 5 > 1$, 5 is a *peak*, so $w_5 = \searrow$. There is one descent pair to the left of 5 which contains 5:

$$8 > 5 > 3.$$

Thus $p_5 = (31 - 2)_5(\sigma) = 1$.

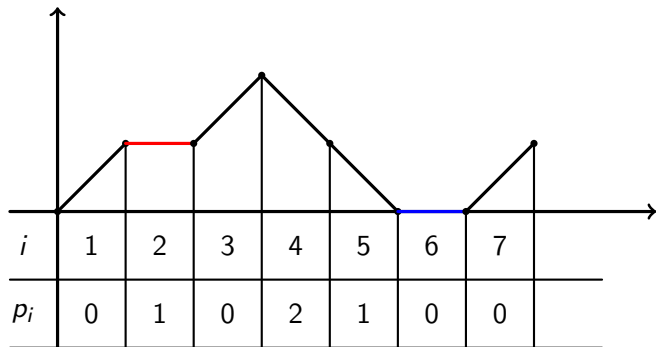
Example: $\Psi_{FV}(\sigma)$, where $\sigma = 697835142$.



Step 6. For $i = 6$. Since $0 < 6 < 9$, 6 is a *double ascent*, so $w_6 = \rightarrow$.
 There is no descent to the left of 6, since 6 is the first letter of σ . Hence

$$p_6 = (31 - 2)_6(\sigma) = 0.$$

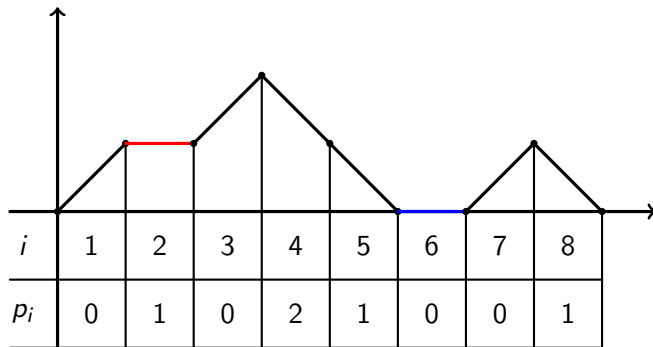
Example: $\Psi_{FV}(\sigma)$, where $\sigma = 697835142$.



Step 7. Look at $i = 7$. Its neighbors are 9 and 8. Since $9 > 7 < 8$, 7 is a *valley*, so $w_7 = \nearrow$. There is no descent to the left of 7 which contains 7. Hence

$$p_7 = (31 - 2)_7(\sigma) = 0.$$

Example: $\Psi_{FV}(\sigma)$, where $\sigma = 697835142$.



Step 8. For $i = 8$. Since $7 < 8 > 3$, 8 is a *peak*, so $w_8 = \searrow$. There is one descent pair to the left of 8 which contains 8:

$$9 > 8 > 7.$$

Thus $p_8 = (31 - 2)_8(\sigma) = 1$.

(p, q) -analogue of generalized Eulerian polynomials

Define

$$A_n(\mathbf{u}, f, g, h, t \mid \alpha, \beta; p, q) := \sum_{\sigma \in \mathfrak{S}_{n+1}} \mathbf{wt}(\sigma) h^{D(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)},$$

where $\mathbf{wt}(\sigma)$ is the weight used in $A_n(\mathbf{u}, f, g, t \mid \alpha, \beta)$ and

$$(31-2)(\sigma) = \sum_{k=1}^n (31-2)_k(\sigma), \quad (2-13)(\sigma) = \sum_{k=1}^n (2-13)_k(\sigma).$$

$$D(\sigma) := \sum_{i \in \text{Pk}(\sigma) \setminus \{n\}} i - \sum_{j \in \text{Val}(\sigma)} j,$$

where $\text{Val}(\sigma)$ and $\text{Pk}(\sigma)$ are the sets of valleys and peaks, respectively.

- Statistics pk val, dd, and da correspond to the number of steps $\searrow$, $\nearrow$, $\rightarrow$, and $\rightarrow$;
- Statistics $(31-2)_i$ and $(2-13)_i$ correspond to the labels p_i and $h_i - p_i$;
- Statistic D corresponds to the area under the associated Motzkin path, and the area of w is given by

$$\text{area}(w) = \sum_{i=1}^n h_i.$$

-

Combining these and the FV-bijection, we have

$$\begin{aligned} & \mathbf{wt}(\sigma) h^{D(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)} \\ &= \prod_{w_i = \nearrow} (u_1 h^{h_i} p^{p_i} q^{h_i - p_i}) \prod_{w_i = \searrow} (u_2 h^{h_i} (\alpha f)^{\chi(p_i=0)} \dots) \prod_{w_i = \rightarrow} (\dots) \prod_{w_i = \rightarrow} (\dots). \end{aligned}$$

Then summing over all $p_i \in \{0, \dots, h_i\}$ for $i \in [n]$

Continued fraction expansion

We define the generalized (p, q) -analogue of a positive integer $k \in \mathbb{N}^*$, with auxiliary parameters α and β , as follows:

$$[k; \alpha, \beta]_{p,q} := (\alpha - 1)p^{k-1} + (\beta - 1)q^{k-1} + \frac{p^k - q^k}{p - q}.$$

In particular, $[k]_{p,q} := [k; 1, 1]_{p,q} = p^{k-1} + p^{k-2}q + p^{k-3}q^2 + \dots + q^{k-1}$.

Theorem (Xu and Zeng)

We have

$$\sum_{n=0}^{\infty} A_n(\mathbf{u}, f, g, h, t \mid \alpha, \beta; p, q) z^n = \mathcal{J}(z; \mathbf{b}, \lambda),$$

where

$$b_k = h^k \left(u_3 [k+1; 1, \alpha t]_{p,q} + u_4 [k+1; \beta t, 1]_{p,q} \right),$$
$$\lambda_k = u_1 u_2 h^{2k-1} [k]_{p,q} [k+1; \beta g, \alpha f]_{p,q}.$$

Application to Gamma-positivity

Considering the specialization

$$\mathbf{u} = (1, xu_2, u_3, xu_4) \text{ and } \beta = \alpha,$$

we obtain ($\text{des} = \text{dd} + \text{val}$)

$$A_n(\mathbf{u}, f, g, h, t \mid \alpha, \alpha; p, q) = \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{des}(\sigma)} u_2^{\text{val}(\sigma)} \dots p^{31-2(\sigma)} q^{2-13(\sigma)}.$$

For $0 \leq k \leq \lfloor n/2 \rfloor$, let $\gamma_{n,k}(x, u_3, u_4, f, g, h \mid t, \alpha \mid p, q)$ be the coefficient of u_2^k in the expansion

$$A_n(\mathbf{u}, f, g, h, t \mid \alpha, \alpha; p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}(x, u_3, u_4, f, g, h \mid t, \alpha; p, q) u_2^k.$$

Consider the two special cases ($\alpha = t = 1$ or $p = q = 1$)

$$A_n(x, u_2, u_3, u_4, f, g, h \mid p, q) := A_n(\mathbf{u}, f, g, h, \mathbf{1} \mid \mathbf{1}, \mathbf{1}; p, q);$$

$$B_n(x, u_2, u_3, u_4, f, g, h \mid t, \alpha) := A_n(\mathbf{u}, f, g, h, t \mid \alpha, \alpha; \mathbf{1}, \mathbf{1}).$$

Theorem (Xu and Zeng)

For $n \geq 0$, the following expansions hold:

$$A_n(x, u_2, u_3, u_4, f, g, h | p, q) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^a(f) (xu_2)^k (u_3 + xu_4)^{n-2k},$$

$$B_n(x, u_2, u_3, u_4, f, g, h | t, \alpha) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_{n,k}^b(f) (xu_2)^k (u_3 + xu_4)^{n-2k},$$

where

$$\gamma_{n,k}^a(f) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=k}^{\text{dd}=0}} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} p^{31-2(\sigma)} q^{2-13(\sigma)},$$

$$\gamma_{n,k}^b(f) = \sum_{\sigma \in \mathfrak{S}_{n+1, \text{des}=k}^{\text{dd}=0}} f^{\text{lmaxpk}(\sigma)-1} g^{\text{rmaxpk}(\sigma)-1} h^{\text{D}(\sigma)} t^{\text{lmaxda}(\sigma)} \alpha^{\text{lmax}(\sigma)+\text{rmax}(\sigma)-2}.$$

When $u_2 = u_3 = u_4 = 1$, we obtain the γ -expansion form. Note that in the case $f = 0$, we obtain analogous results for the binomial-Eulerian polynomials.

Sketch of the proof

After the substitutions

$$u_2 \rightarrow \frac{u_2(u_3 + xu_4)^2}{x}, \quad z \rightarrow \frac{z}{u_3 + xu_4},$$

we obtain

$$\sum_{n \geq 0} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\gamma_{n,k}(x, u_3, u_4, f, g, h \mid t, \alpha; p, q)}{x^k (u_3 + xu_4)^{n-2k}} u_2^k z^n = \mathcal{J} \left(z; \frac{b_k}{u_3 + xu_4}, \lambda_k \right),$$

where

$$b_k = h^k \left(u_3[k+1; \alpha t, 1]_{p,q} + xu_4[k+1; 1, \alpha t]_{p,q} \right),$$
$$\lambda_{k+1} = h^{2k+1} u_2[k+1]_{p,q} [k+2; \alpha g, \alpha f]_{p,q} \quad (k \geq 0).$$

Hence, the desired γ -expansion follows once

$$\frac{b_k}{u_3 + xu_4}$$

is a polynomial with nonnegative coefficients.

Sketch of the proof

The key quotient simplifies in two important cases.

When $\alpha = t = 1$, for $k \geq 0$,

$$\frac{b_k}{u_3 + xu_4} = h^k[k+1]_{p,q}.$$

Thus the coefficient of $u_2^k z^n$ is independent of x, u_3, u_4 . Setting $x = u_3 = 1$ and $u_4 = 0$, we get

$$\frac{\gamma_{n,k}(x, u_3, u_4, f, g, h \mid 1, 1; p, q)}{x^k(u_3 + xu_4)^{n-2k}} = \gamma_{n,k}^a(f, g, h \mid p, q).$$

Similarly, when $p = q = 1$, for $k \geq 0$,

$$\frac{b_k}{u_3 + xu_4} = h^k(k + \alpha t).$$

Therefore the required γ -positivity follows in these two cases.

Further γ -coefficient factorization

Theorem (Xu and Zeng)

There exist polynomials $E_{n,k}(h | p, q) \in \mathbb{N}[h, p, q]$ and $E_{n,k}(f, h | t, \alpha) \in \mathbb{N}[f, h, t, \alpha]$ such that for $0 \leq k \leq \lfloor n/2 \rfloor$,

$$\gamma_{n,k}^a(1, 1, h | p, q) = (p + q)^k E_{n,k}(h | p, q),$$

$$\gamma_{n,k}^b(f, f, h | t, \alpha) = 2^k E_{n,k}(f, h | t, \alpha)$$

The above result follows from

$$\frac{b_k}{u_3 + xu_4} = h^k [k+1]_{p,q} \in \mathbb{N}[h, p, q], \quad \frac{\lambda_k}{p+q} = h^{k-1} \frac{[k+1]_{p,q} [k+2]_{p,q}}{p+q} \in \mathbb{N}[h, p, q];$$
$$\frac{b_k}{u_3 + xu_4} = h^k (k + \alpha t) \in \mathbb{N}[h, \alpha, t], \quad \frac{\lambda_k}{2} = h^{k-1} \frac{(k+1)(k+2\alpha f)}{2} \in \mathbb{N}[h, f, \alpha].$$

We give a combinatorial interpretations for $E_{n,k}(h | p, q)$ and $E_{n,k}(f, h | t, \alpha)$ based on André permutaions (type I and II) using group actions

Bi-Stirling-Euler-Mahonian polynomials

q -derivative, Carlitz-type rational function

Bi-Stirling Eulerian numbers

The bi-Stirling Eulerian numbers $A(r, s | \alpha, \beta)$ by

$$\sum_{r,s=0}^{\infty} A(r, s | \alpha, \beta) \frac{x^r y^s}{(r+s)!} = (1 + xF(x, y))^{\alpha} (1 + yF(x, y))^{\beta}, \text{ with } F(x, y) = \frac{e^x - e^y}{xe^y - ye^x}.$$

Carlitz and Scoville (1974) proved the recurrence

$$A(r, s | \alpha, \beta) = (s + \alpha)A(r - 1, s | \alpha, \beta) + (r + \beta)A(r, s - 1 | \alpha, \beta)$$

and also found the combinatorial interpretation

$$A(r, s | \alpha, \beta) = \sum_{\substack{\sigma \in \mathfrak{S}_{r+s+1} \\ \text{des}(\sigma) = r}} \alpha^{\text{lmin}(\sigma)-1} \beta^{\text{rmin}(\sigma)-1}.$$

Bi-Stirling-Euler-Mahonian numbers

Define q -analogue of n and n factorial

$$[n]_q := 1 + q + q^2 + \cdots + q^{n-1}$$

$$[n]_q! := [n]_q [n-1]_q \cdots [1]_q.$$

Bi-Stirling-Euler-Mahonian numbers

Gaudin (2024) noticed that $A(r, s | \alpha, \beta)$ is a rescaled version of a subfamily of the mixed Eulerian numbers, introduced by Nadeau-Tewari (2021), which led him to propose a new q -analogue $A(r, s | \alpha, \beta)_q$ in the framework of the remixed Eulerian numbers as

$$A(r, s | \alpha, \beta)_q = \frac{1}{[\alpha + \beta - 1]!} A_{(0^r, 1^{\alpha-1}, r+s+1, 1^{\beta-1}, 0^s)}(q)$$

and derived the following q -analog:

$$A(r, s | \alpha, \beta)_q = q^{r+\beta-1} [s + \alpha] A(r-1, s | \alpha, \beta)_q + [r + \beta] A(r, s-1 | \alpha, \beta)_q.$$

Note that when $\beta = 0$, this recurrence appeared in Haglund-Bulter's work on q -rook theory. We define the *bi-Stirling-Euler-Mahonian numbers* $E_{n,k}(\alpha, \beta, q)$ by

$$E_{r+s,r}(\alpha, \beta, q) := A(r, s | \alpha, \beta)_q$$

Mahonian statistics

The *major index* of a permutation $\sigma \in \mathfrak{S}_n$ is defined as the sum of the descents of σ , i.e.

$$\text{maj}(\sigma) = \sum_{\sigma_i > \sigma_{i+1}} i,$$

$\mathfrak{S}_3$	maj
123	0
132	2
213	1
231	2
312	1
321	3

$$\begin{aligned} \sum_{\sigma \in \mathfrak{S}_3} q^{\text{maj}(\sigma)} \\ &= 1 + 2q + 2q^2 + q^3 \\ &= (1 + q + q^2)(1 + q) \end{aligned}$$

Theorem (MacMahon 1915)

$$\sum_{\sigma \in \mathfrak{S}_n} q^{\text{maj}(\sigma)} = [n]_q!$$

Bi-Stirling-Euler-Mahonian polynomials

For $\sigma \in \mathfrak{S}_n$ define the (α, β) -*major index*

$$\widetilde{\text{maj}}(\sigma) = (n - \text{rmin}(\sigma))(\beta - 1) + (\text{des}(\sigma) - \text{lmin}(\sigma) + 1)(\alpha - 1) + \text{maj}(\sigma).$$

Theorem (Xu and Zeng)

For $n \geq 1$, we have

$$\frac{E_n(x \mid \alpha, \beta, q)}{(x; q)_{n+\alpha+\beta}} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n,$$

where

$$E_n(x \mid \alpha, \beta, q) = \sum_{k=0}^n E_{n,k}(\alpha, \beta, q) x^k = \sum_{\sigma \in \mathfrak{S}_{n+1}} x^{\text{des}(\sigma)} [\alpha]^{\text{lmin}(\sigma)-1} [\beta]^{\text{rmin}(\sigma)-1} q^{\widetilde{\text{maj}}(\sigma)}.$$

We prove it by induction and q -calculus.

Further direction: Cluster parking function

quasi-Stirling permutation, Multiset Eulerian polynomials

Further direction: quasi-Stirling permutations

- For the multiset $M = \{1^{k_1}, 2^{k_2}, \dots, n^{k_n}\}$. [MacMahon \(1913\)](#) introduced the multiset Eulerian polynomial

$$A_M(t) = \sum_{\pi \in \mathfrak{S}_M} t^{\text{des}(\pi)}.$$

- A permutation of a multiset $\{1^2, 2^2, \dots, n^2\}$ is called a quasi-Stirling permutation if it avoids the patterns 1212 and 2121.
- [Archer et al. \(2019\)](#) showed that

$$|\overline{\mathcal{Q}}_n| = n! C_n, \quad \text{where} \quad C_n = \frac{1}{n+1} \binom{2n}{n}$$

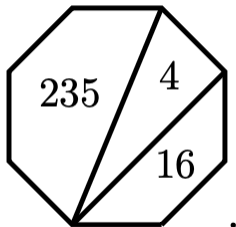
is the n th Catalan number.

- [Elizalde \(2021\)](#) introduced the quasi-Stirling polynomial

$$\overline{Q}_n(t) = \sum_{\pi \in \overline{\mathcal{Q}}_n} t^{\text{des}(\pi)}.$$

Further direction: cluster parking functions

Josuat-Vergès et al. (2025) introduced the cluster parking function complex CPF_n .



A cluster parking function is a labeled face of the cluster complex: each inner $(k + 2)$ -gon carries exactly k labels.

Further direction: cluster parking functions

A key result is that its h -polynomial is the quasi-Stirling polynomial:

$$h(\text{CPF}_n; t) = \overline{Q}_n(t).$$

From the above results, we obtain the chain of equalities

$$\sum_{\pi \in \mathfrak{S}_n} t^{\text{des}(\pi)} (n+1)^{\text{lmin}(\pi)-1} = \overline{Q}_n(t) = A_{\mathcal{M}'}(t) = h(\text{CPF}_n; t),$$

where $\mathcal{M}'$ is any element in the set

$$\left\{ \{1^{n+1}, 2, 3, \dots, n\}, \{1, 2^{n+1}, 3, \dots, n\}, \dots, \{1, 2, 3, \dots, n^{n+1}\} \right\}.$$

Merci!